



PROFESSIONAL CERTIFICATION

MasterCam Training

The MasterCam Course covers CAM for CNC manufacturing. Participants learn 2D/3D toolpaths simulation and post-processing. Training includes mill lathe and multi-axis programming for precision machining in aerospace automotive and mold making.



30-35

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

MasterCam Course

It is the tool to learn CAM programming with MasterCam.

Course Description

This course teaches MasterCam for CNC programmers. Master high-speed machining dynamic milling and multi-axis strategies. Projects cover complex parts tooling and verification for efficient manufacturing in various industries.

Duration

30-35

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Master MasterCam interface.
- ✓ Create 2D toolpaths.
- ✓ Program 3D surfaces.
- ✓ Simulate machining processes.
- ✓ Optimize high-speed cuts.
- ✓ Handle multi-axis ops.
- ✓ Generate post codes.

Target Audience

- CAM Engineers programming.
- Fabrication pros in CNC.
- Production manufacturing.

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic machining knowledge needed.
- Computer with MasterCam installed.
- Interest in CAM tools.

MasterCam Course

Start with MasterCam basics and mill programming.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
30 Hrs

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
33 Hrs

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
35 Hrs

MasterCam Course

Start with MasterCam basics and mill programming.

01 Navigating the Mastercam User Interface

- > Explore ribbon tabs managers and graphics window
- > Customize Quick Access Toolbar and keyboard shortcuts
- > Manage files configurations and system settings
- > Utilize selection tools AutoCursor and Quick Masks
- > Access help resources tutorials and status bar information

02 Creating and Editing 2D Wireframe Geometry

- > Draw lines arcs circles and splines precisely
- > Apply fillets chamfers offsets and trim operations
- > Use mirroring rotating scaling and translating tools
- > Chain geometry for toolpath selection effectively
- > Import and repair external CAD files like DXF/IGES

03 Working with Levels Planes and Views

- > Organize entities using Levels Manager
- > Create and manage custom planes and Cplanes
- > Set up viewsheets and dynamic gnomon controls
- > Transform geometry across different planes
- > Control visibility and attributes for complex parts

04 Building Solid Models and Surfaces

- > Create extrude revolve sweep and loft solids
- > Generate ruled lofted and swept surfaces
- > Combine trim and stitch surface geometry
- > Convert wireframe to solids or surfaces
- > Analyze and repair imported model geometry

05 Setting Up Machine Groups and Stock Models

- > Define machine definitions and control definitions
- > Configure stock setup and workholding fixtures
- > Assign tools and create tool libraries
- > Establish coordinate systems and WCS origins
- > Prepare multiple setups within one file

06 Generating 2D Milling Toolpaths Effectively

- > Create contour pocket face and drill operations
- > Apply chaining lead-in/out and ramping strategies
- > Set depths stepovers and multiple passes
- > Use dynamic motion for efficient roughing
- > Verify toolpaths with backplot and simulation

MasterCam Course

Start with MasterCam basics and mill programming.

07 Programming 3D Roughing and Semi-Finishing Toolpaths

- > Apply roughing strategies like area roughing and core roughing
- > Use rest roughing and leftover material detection
- > Generate semi-finishing paths for smooth transitions
- > Control tool engagement and scallop heights
- > Optimize feeds speeds and toolpath filtering

08 Creating 3D Finishing Toolpaths

- > Implement parallel contour scallop and horizontal finishing
- > Apply flowline and raster strategies for complex surfaces
- > Use pencil and leftover finishing techniques
- > Adjust stepover tolerances and boundaries
- > Achieve high-quality surface finishes efficiently

09 Drilling and Hole-Making Operations

- > Define drill cycles including peck chip break and tap
- > Create hole patterns with grid and circle options
- > Apply custom drill depths and retracts
- > Group and sort drilling operations logically
- > Verify drilling paths with simulation tools

10 Working with Lathe and Turning Toolpaths

- > Set up lathe machine groups and stock
- > Generate facing roughing finishing and grooving paths
- > Apply threading boring and parting operations
- > Use canned cycles and tool nose radius compensation
- > Simulate turning processes for accuracy

11 Implementing Multi-Axis and 4/5-Axis Machining

- > Configure rotary axis and indexing setups
- > Create curve/5-axis swarf and flow toolpaths
- > Program port machining and multi-surface strategies
- > Define tilt lead/lag and collision avoidance
- > Verify multi-axis motion for safe machining

12 Tool Management and Library Optimization

- > Build and organize tool libraries effectively
- > Define tool geometries holders and parameters
- > Assign feeds/speeds from material databases
- > Use toolpath templates for consistency
- > Manage tool wear and life tracking

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13 Toolpath Verification and Simulation Techniques

- › Perform backplot and 2D/3D verification
- › Use Mastercam Simulator for stock removal display
- › Detect collisions gouges and rapid errors
- › Analyze cycle times and material removal rates
- › Validate toolpaths before posting

14 Post-Processing and NC Code Generation

- › Select and customize post processors
- › Generate G-code for specific CNC controls
- › Edit and review NC output files
- › Use Code Expert for code analysis
- › Set up posting parameters and comments

15 Optimizing Toolpaths for Production Efficiency

- › Apply high-efficiency strategies and trochoidal paths
- › Use feature-based machining and templates
- › Implement nesting for multiple parts
- › Reduce air cuts and optimize entry/exit
- › Refine parameters for reduced cycle times

16 Advanced Machining Workflows and Best Practices

- › Develop complete programs for complex parts
- › Integrate hybrid milling/turning operations
- › Prepare documentation with setup sheets
- › Apply industry-specific strategies for molds/dies
- › Export and archive production-ready files

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

Early Bird Discount

Free

Demo Session



Call Us Now

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